WEEK 1 · LESSON 6 OF 24 · HANDOUT

Week 1 Project: Analyse an AI System

Study time: **about 60 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=3AZF1TTDdEM — open the lesson online to watch inside the course.

🎓 **Go deeper:** Harvard CS50x 2026: Artificial Intelligence — full lecture — https://www.youtube.com/watch?v=-9bo8HlSxwQ

Your goal in this lesson: Bring together everything from Week 1 by analysing one real AI system: where it sits on the AI map, its PEAS description, its environment, its problem type and its history.

What you will do

This video shows Zipline's drone deliveries in Rwanda — a real AI-powered system working in our hills and weather. Use it as a model, then analyse a system of your own choice. Your analysis should be about **two pages** (or 8–10 slides). It is the kind of short, clear report companies expect from a junior analyst.

Step 1 — Choose your system

Choose one AI system you can learn about. Ideas:

- Drone delivery (Zipline)
- Mobile-money fraud detection
- A crop-disease app that reads leaf photos
- A Kinyarwanda speech or translation tool
- A chatbot used by a bank, telecom or government service
- A recommendation feed (YouTube, TikTok, Jumia)

Step 2 — Place it on the AI map

- Is it rule-based AI, machine learning, deep learning — or a mix? Give evidence.
- Is it narrow AI? (It is — but explain what its narrow task is.)

Step 3 — Write its PEAS table

	Your system
Performance measure	
Environment	
Actuators	
Sensors	

Then name **one risk** if the performance measure were chosen badly.

Step 4 — Classify its environment

Fully or partially observable? Deterministic or stochastic? Episodic or sequential? Static or dynamic? Single- or multi-agent? One sentence of reasoning for each.

Step 5 — Name the problem type(s)

Is it classification, regression, clustering, recommendation, anomaly detection, generation or sequential decision-making? Many systems combine several — a drone system *plans routes* (search, sequential decisions) and may *predict* weather risk (regression or classification).

Step 6 — History and impact

- Which of the three ingredients (data, computing, methods) made this system possible?
- Who benefits? Who could be harmed or left out?

Checklist before you submit

- Every section above is answered.
- At least one source is named (website, article or video).
- You used the correct terms: agent, PEAS, environment properties, problem type.
- Simple, clear sentences — a manager should understand it in five minutes.

Remember this: You can now describe any AI system precisely, using the same language professionals use. This skill is the base for everything else in the course.

Name: _____ Date: _____

Activity

Task 1

Complete the six steps for your chosen system. If you are studying with friends, swap reports and check each other's PEAS tables and problem types. Keep your report — it becomes the first page of your final project in Week 4.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. In the Week 1 project, what should the PEAS table describe?
   - A. The programmer's CV
   - B. Performance measure, Environment, Actuators and Sensors of the system
   - C. Only the sensors
   - D. The price of the system
2. A drone delivery system plans routes. Which problem type is this mainly?
   - A. Sequential decision-making / search
   - B. Recommendation
   - C. Generation of images
   - D. Clustering
3. Which ingredient question belongs in the history and impact step?
   - A. How many employees the company has
   - B. Which data, computing power and methods made the system possible
   - C. What the CEO eats
   - D. Which colour the logo is
4. Why name at least one source in your report?
   - A. To make the report longer
   - B. So readers can check your facts
   - C. It is not needed
   - D. Because it is decoration
5. A system uses both hard rules and a learned model. How should you place it on the AI map?
   - A. As a mix, with evidence for each part
   - B. Only deep learning
   - C. Only rule-based
   - D. Not AI
6. What is a good risk to name about the performance measure?
   - A. Rewarding only speed could make the system unsafe
   - B. The app icon is small
   - C. Users like the colour blue
   - D. The website loads slowly

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